TTA Results: op100 → Re500 Kolmogorov Flow

2026-06-23

Setting. FNO pretrained at Re100 (`op100`, $n_{\text{modes}} = 8$, 128^2), adapted label-free to Re500 at inference via physics-residual TTA ($\mathcal{L} = \text{pde} + 5 \cdot \text{ic}$, oracle ν). Metric: held-out $k \leq 7$ relL2 vs GT, time-split early / late. Reference ceiling: op500 (supervised, same arch) late 0.473.

---

1. The main finding

Physics-residual TTA **solves early, hard-walls late** — and the wall is real:

	early $k \leq 7$	late $k \leq 7$
op100 source (baseline)	0.271	0.678
op300 best free ID op	0.234	0.527
op500 supervised ceiling	**0.248**	**0.473**
TTA best (N=160, lr=3e-4)	**0.175**	**0.581**

Early is below op500 supervised and monotone in data size. Late plateaus at 0.58–0.59 for $N \geq 120$, $\sigma(\text{seeds}) \approx 0.001$. A band-limited $k \leq 7$ residual objective (Exp#1) moves late by −0.022 only (0.578 → 0.556).

The wall is invariant to every lever tried on the instantaneous objective: data size, learning rate, seed, and residual band.

---

2. What the wall is not

Five axes ruled out:

Capacity (spatial, temporal, architectural). Modes $n_8 \to n_{32}$, width $h_{32} \to h_{128}$: late $\Delta \leq 0.017$. GT projected onto $K_t = 8$ time-modes holds at 0.135 ≪ wall 0.55 — the GT late field lives in 8 modes; the model error does not. UNet with a single spatial-FNO bottleneck reaches the FNO late wall at both 128^2 (0.478 vs 0.472) and 256^2 (0.577 vs 0.571): the wall is architecture-invariant.

Chaos / IC-unpredictability. Gate 2 (phase-spread capstone): ensemble phase dispersion $D_{\text{late}} = 0.07$–0.30, growing ≈ 1.3×, does not saturate toward 1 even at perturbation $\delta = 0.30$ exceeding the operator's early error. Late phase is learnable, not intrinsically chaotic. Paired op-quality at 256^2 ($n = 100$): op500 > op300 late 61/100 ($p = 0.035$), not 50/50 — the wall has a systematic reducible component.

Self-consistency as a trainable surrogate. Self-inconsistency rank-tracks GT late (Spearman 0.40–0.65) but is non-causal: halving the inconsistency leaves GT late flat or worse. Transductive dose-response: inconsistency ↓ on 100% of instances, GT late ↑ on 100%. Oracle distillation with data anchoring saturates at the supervised ceiling ≈ 0.48–0.52, not below it.

---

3. What the error looks like

Per-mode decomposition: $|\hat{u} - \hat{g}|^2 = (a_u - a_g)^2 + 2a_u a_g(1 - \cos\Delta\phi)$. Collapse-independent per-shell: $R_k = \Sigma|\hat{u}|^2/\Sigma|\hat{g}|^2$ (energy ratio), A_k = GT-energy-weighted phase cosine.

Both R_k and A_k fall monotonically with k — they co-degrade, not independently:

k	R_k (OOD 128^2)	A_k (OOD 128^2)
1	0.83	1.00
3	0.54	0.84
5	0.33	0.56
7	0.16	0.26

Aggregate $k \leq 7$: amplitude-leaning (relL2 amp 0.281 > phase 0.243, op500). Phase fraction grows with wavenumber and with time (early → late at every shell). Phase concentrated $k = 2$–5 (fraction 0.40–0.59); $k = 0$–1 purely amplitude.

One mechanism: low-pass hedging under positional uncertainty. Phase-blind magnitude correction raises late error (0.581 → 0.631) — energy cannot be fixed while phase is wrong. Loss reweighting (time axis, band equalization) moves late < 0.006 and non-monotone.

---

4. The one lever that works

Oracle re-anchoring — restart from a true GT mid-trajectory field every s frames — breaks the wall **on 100% of instances** across all operators:

operator	oneshot late	oracle late	Δ
op100, 128^2, stride 16	0.678	**0.361**	−0.310
op500, 128^2, stride 16	0.473	**0.283**	−0.185
fno16, 256^2, stride 16	0.571	**0.230**	−0.328

The recoverable signal lives entirely in $k \leq 7$, concentrated $k = 2$–7 (closure gate kc-sweep). Correct $k > 7$ buys nothing: hi_oracle ≈ model at $k_c = 7$. Solver closure confirms this is physics: true-solver low-k divergence from 100%-corrupted high-k = 0.06–0.12 ≪ wall 0.47.

Own-field restart compounds error for far-OOD ops (op100 model: 0.912–0.771, all worse than one-shot 0.678). fno16 own-field is modestly positive (80–90% win) due to lower early error (0.164 vs 0.279), but oracle gap ≈ 0.27 remains.

The gap between oracle (0.230) and own-field (0.525) restart is 0.295 — the remaining addressable loss.

---

5. Proposed direction

Self-consistency cannot close the oracle gap (non-causal). The missing piece is a training objective that lets the model see correct mid-field phase states.

Multi-step rollout training with segmented GT anchoring. Train the 256^2 FNO from mid-trajectory GT snapshots at stride $s \in \{8, 16, 32\}$, not only from IC. Each segment starts from a true field and is supervised T' steps forward (data + pde loss). Training-objective change only — no architecture change, no label requirement at inference.

Mechanistic basis: given a recent true field, the operator already predicts far below the one-shot wall (fno16: 0.571 → 0.230). The wall is stale-IC accumulation that the one-shot objective never penalises. Segmented training exposes the model to correct phase-state anchors at multiple trajectory points, directly targeting the low-pass hedging mechanism.

Discriminator: late $k \leq 7 < 0.47$ without oracle access at inference confirms trajectory accumulation as the mechanism. If the wall persists, the failure mode is the Re-shift domain gap rather than trajectory length — testable with a segmented-start arm on Re500 pool directly.